

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE – RAIGAD -402 103
Winter Semester Examination – December - 2019

Branch: Electrical Engineering

Sem.:- V

Subject: -Power plant Engineering (BTEEOE506B)

Marks: 60

Date:18/12/2019 -

Time: - 3 Hr.

Instructions to the Students

1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly

- | | (Marks) |
|--|----------------|
| Q.1. a) Explain Indian energy scenario in detail? | (6) |
| b) Explain methods and types of tariff for electrical energy in power plant economics? | (6) |
| Q.2. a) Draw the schematic layout of a modern thermal power plant and explain the functions of each component and operation of plant? | (6) |
| b) Classify the different type of reactors and explain pressurized water reactor (PWR) power plant? | (6) |
| Q.3. a) Draw the schematic layout of gas turbine power plant and its type also explain the functions of each component and operation of plant? | (6) |
| b) Pelton wheel of 3m runner work under head of 800m if it runs at 60rpm. The discharge rate in the runner is $3\text{m}^3/\text{s}$. Find:
i) Input power of runner.
ii) Shaft power from runner having mechanical efficiency (η_m) of 92%.
iii) Net power output if generator is 96% efficient.
iv) Specific speed of turbine.
v) Number of turbines needed to generate 100MW power. | (6) |
| Q.4. a) Draw the schematic layout of diesel power plant and explain the functions of each component and operation of plant? | (6) |

b) A two stroke engine develops 400kW output with mechanical efficiency of 0.82. The rate of fuel consumption is 2.4kg/min and air –fuel ratio used in the engine is 18:1. the heating value of the fuel is 41500 kJ/kg.

find the followings: i) Indicated power

ii) Friction power

iii) Break thermal efficiency

iv) Indicated thermal efficiency

v) Fuel and air consumption per hour

(6)

Q.5. a) Explain in detail Binary vapour cycle?

(6)

b) A generating unit of 10MW capacity supplies following loads:

Domestic consumers with a maximum demand 6MW at load factor of 20%

Small industrial load with a maximum demand 3.6MW at load factor of 50%

Street light load with a maximum demand 400kW at 30% load factor.

Find: overall cost of energy for kWh each type of consumers using

following data: capital cost of the plant = Rs.10,000/kW

Total Running cost = Rs. 36, 00,000/year

Annual rate of interest and depreciation on capital cost = 10%

(6)

Q.6. a) Explain concept of grid and condition to interface different power plant to grid?

(6)

b) Write a short on followings:

(6)

i) Parallel operation of various generating sources and load sharing.

ii) Topology of grid

****PAPER END****